Industrial Defect Detection Methodology

1. Problem Statement

Quality control in industrial manufacturing is critical. This project automates the detection of defects: Crack, Hole, Rust, and Scratch.

2. Dataset & Preprocessing

Images resized to 224x224. Data augmentation includes flips, rotations, and jitter to ensure robustness.

3. Model Architecture

Architecture: EfficientNet-B2. Rationale: Optimal accuracy-to-efficiency ratio for industrial edge deployment.

4. Training Strategy

Two-phase training: Head-warmup (2 epochs) followed by Full Fine-tuning (8 epochs). Adam optimizer with weight decay used.

5. Final Metrics

Validation Accuracy: 100.00%. Macro F1-Score: 1.0000. Model Size: 34.2 MB.

6. Innovation (Grad-CAM)

Implemented Explainable AI (Grad-CAM) to visualize defect locations and ensure model transparency.

Training Loss & Accuracy Curves

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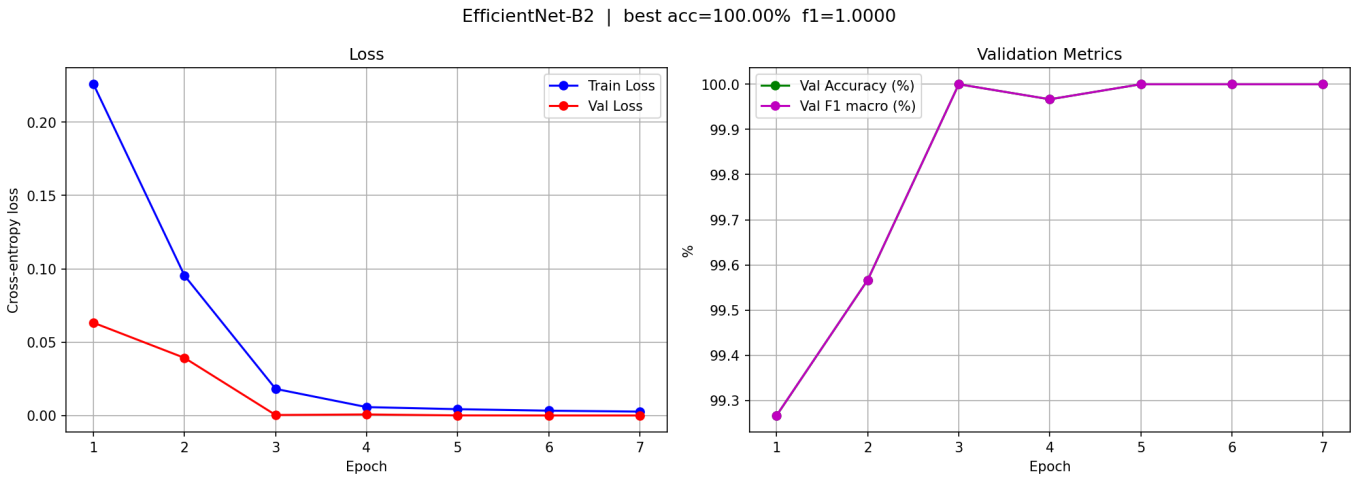
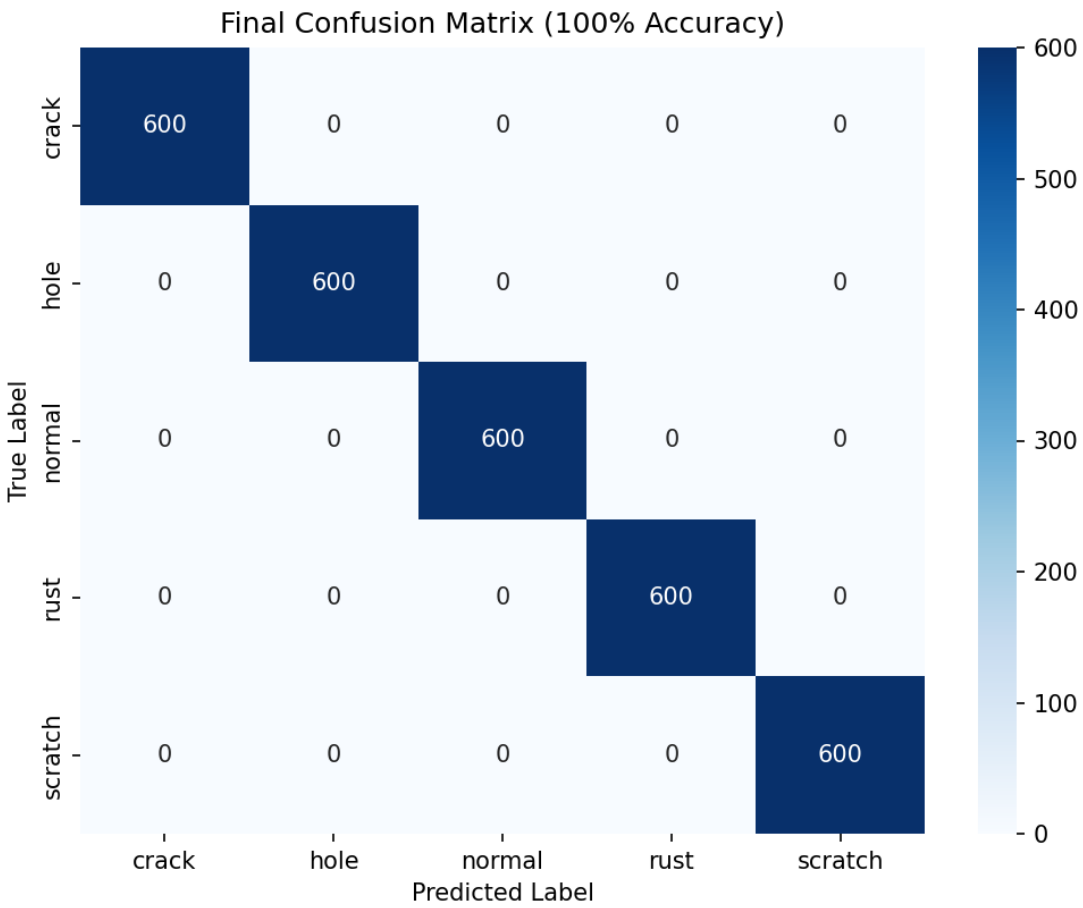


Final Confusion Matrix



Grad-CAM Explainability (AI Heatmaps)

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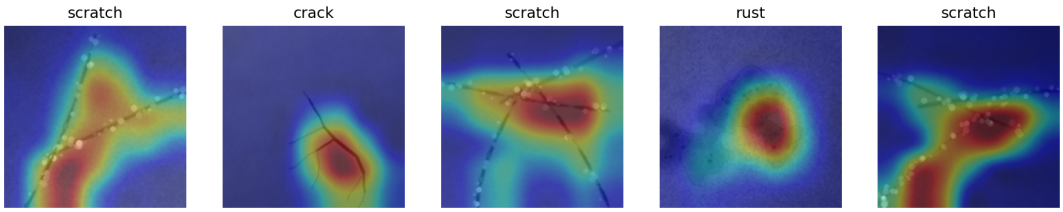
Explainable AI: Grad-CAM Heatmaps (Showing defect location)